INCH-POUND

MIL-I-26382C(USAF)
01 January 1994
MIL-I-26382B (USAF)
25 March 1985

MILITARY SPECIFICATION

* INDICATOR SET, LIQUID OXYGEN QUANTITY A/A24J-4 AND A/A24J-8

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of 10 liter liquid oxygen quantity indicator set designated by the following types:

a. Type A/A24J-4
b. Type A/A24J-8

2. APPLICABLE DOCUMENTS

2.1 Government Documents

2.1.1 Specification and Standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-P-116	Preservation, Method of
MIL-L-25467	Lighting, Integral, Red, Aircraft Instrument, General Specification For
MIL-I-25645	Indicator, Liquid Oxygen Quantity, Capacitance Type, General Specification For

STANDARDS

FEDERAL

FED-STD-595	Colors

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Logistics Services Division, SA-ALC/TILDD, Bldg 171, Post C-12, 485 Quentin Roosevelt Rd, Kelly AFB, TX 78241-6425 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6680

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MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-781	Reliability Test Exponential Distribution
MIL-STD-831	Test Reports, Preparation of
MIL-STD-2073-1	DOD Material Procedures for Development and Application of Packaging Requirements (Part 1 of 2 Parts)
MS33558	Numeral and Letter, Aircraft Instrument Dial, Standard Form Of

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents cited in the solicitation not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-D-3951	Packing, Commercial

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

2.3 Order of Precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Indicator sets furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.3).

3.2 General specification. The requirements of MIL-I-25645 shall apply as requirements of this specification with the exceptions and additions called out herein. When the two specifications conflict this specification shall govern.

3.3 Configuration. The indicator set shall consist of a master indicator and a repeater indicator. These two indicators shall include all of the equipment necessary (except the cables) to meet the performance requirements specified herein and in MIL-I-25645.

3.4 Cases. The length of the case for the master indicator shall not exceed 5.0 inches. The length of the case for the repeater indicator shall not exceed 3.5 inches.

3.5 Dials. The dials for both the master indicator and the repeater indicator shall conform to figure 1.

3.6 Electrical connectors. Electrical connectors shall be in accordance with figures 2 and 3. The master indicator and the repeater indicator shall be designed internally so that they will function properly when the mating connectors are wired as shown in figures 2 and 3.

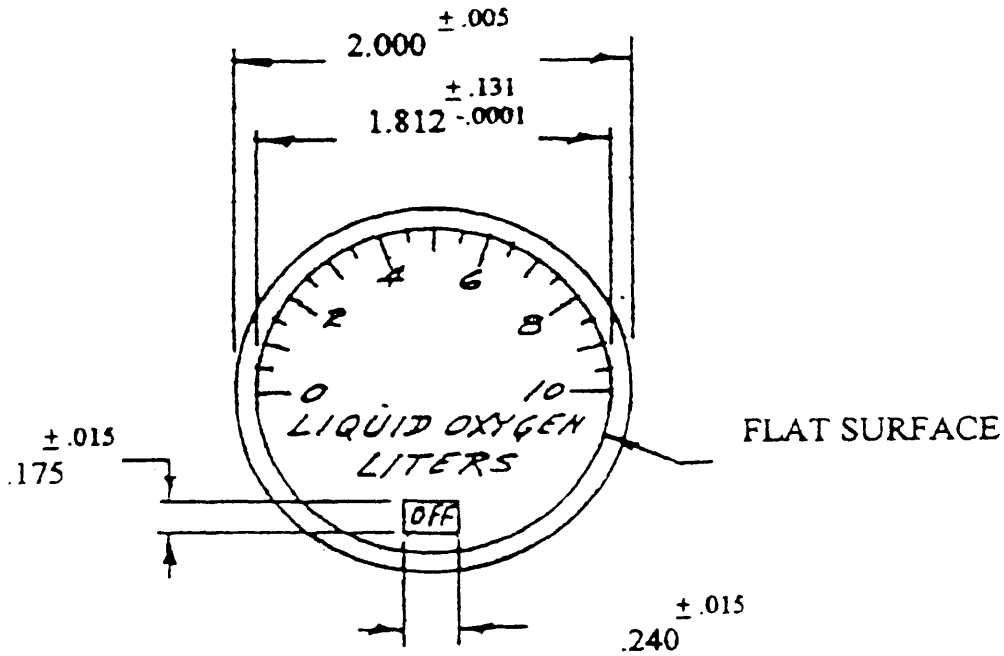
3.7 Capacitance. For design purposes, the capacitance for zero reading shall be 123.5 picofarad (pf).

3.8 Adjustments. The zero calibration adjustment for both the master indicator and the repeater indicator shall be capable of adjusting the pointer to zero indication over an input capacitance range (to the master indicator) of at least 120.0 pf to 135.0 pf. With the indicator adjusted to zero indication at 120.0 pf input capacitance, the indicator shall be capable of calibration to full

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scale indication at 178.0 pf input capacitance by means of the full calibration adjustment. With the indicator adjusted to zero indication at 135.0 pf input capacitance, the indicator shall be capable of calibration to full scale indication at 193.0 pf input capacitance by means of the full calibration adjustment.

3.9 Weight. The weight of the master indicator shall not exceed 1.5 pounds. The weight of the repeater indicator shall not exceed 0.75 pounds.



NUMERALS & LETTERS SHALL CONFORM TO MS33558

MARKINGS	HEIGHT OR LENGTH +/- .010	WIDTH OF LINE OR GRADUATION +/- .005	FINISH COLOR PER FED STD 595
NUMERALS	.140	.025	
LIQUID OXYGEN	.140	.020	
MAJOR GRAD	.188	.031	37875
MINOR GRAD	.125	.015	
SUB MINOR	.080	.015	
LITERS	.100	.015	
BACKGROUND OF DIAL			37038
OFF	.094	.015	37875
OFF FLAG BACKGROUND			31158

UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES

Figure - 1 DIAL

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MASTER INDICATOR CONNECTOR (AMPHENOL PN 182-2-1000 EXCEPT FACE SEAL IN ACCORDANCE WITH PN 67-2058-1862, OR EQUAL)

SHIELDED LEAD TO INPUT	CO AX	AX
REPEATER - PIN A	A	A
115 VOLTS, 400 CYCLE	B	B
LOW LEVEL WARNING LIGHT	C	C
REPEATER - PIN D	D	D
5 VOLTS, 400 CYCLE	E	E
PRESS-TO-TEST	F	F
UNSHIELDED LEAD TO INPUT	H	H
GROUND	J	J
LOW LEVEL WARNING LIGHT	K	K
REPEATER - PIN L	L	L
REPEATER - PIN M	M	M
MATING CONNECTOR ↗		

(PN 165-62 MANUFACTURED BY AMPHENOL CONNECTOR DIVISION, 2801 S. 25TH AVE, BROADVIEW, ILLINOIS, 60153, OR EQUAL)

Figure 2 MASTER INDICATOR CONNECTOR WIRING

REPEATER INDICATOR CONNECTOR (AMPHENOL P/N 182-11 OR EQUAL)		↓
MASTER - PIN A	A	A
NO CONNECTION	B	B
MASTER - PIN D	D	D
5 VOLT, 400 CYCLE	E	E
GROUND	J	J
MASTER - PIN L	L	L
MASTER - PIN M	M	M
MATING CONNECTOR (PN 165-10 MANUFACTURED BY AMPHENOL CONNECTOR DIVISION, 2801 S. 25TH AVE, BROADVIEW, ILLINOIS 60153, OR EQUAL)	↑	

Figure 3 REPEATER INDICATOR CONNECTOR WIRING

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3.10 Accuracy. When the indicator set is tested as specified in MIL-I-25645, the accuracy of the master indicator shall be in accordance with table I. the tolerance for room temperature reading is the allowable deviation from the nominal calculated values. Repeater tolerance is the allowable deviation from the corresponding master indicator reading. All other tolerances are the allowable deviation from the actual room temperature readings.

3.11 Power supply variation. Variations of supply voltage from 105 volts to 125 volts and frequency of 320 to 480 cycles per second shall not cause indicator errors, exceeding those specified in table I, when the indicator is tested as specified in MIL-I-25645. For the 105 volt test point, the indicator shall be checked at frequencies of 320 and 480 cycles per second. For the 125 volt test point, the indicator shall be checked at frequencies of 320 and 480 cycles per second.

TABLE I. TOLERANCES FOR CAPACITANCE READINGS.

Test	Tolerance Plus of Minus
Accuracy - room temperature	1.0 pf
Low temperature operation	1.0 pf
High temperature operation	1.0 pf
Vibration	0.6 pf
Power supply variation	1.2 pf
Low temperature storage and operation	1.0 pf
High temperature storage and operation for Type A/24J-4	1.0 pf
High temperature storage and operation for Type A/A24J-8	0.6 pf
Salt spray	0.6 pf
Cycling	0.6 pf
Repeater indicator	0.6 pf

3.12 Dead spot. The dead spot shall not exceed 1.0 percent of full scale indication.

3.13 Integral lighting. The master indicator and the repeater indicator shall have integral lighting in accordance with MIL-L-25467 for type A/24J-4 and MIL-L-27160 for Type A/A24J-8.

3.14 Power-failure flag. Both the master indicator and the repeater indicator shall be provided with a power-failure flag. The flag shall be blank for power-on conditions and shall indicate "OFF" for power-off conditions.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract of purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.2 Sampling, inspecting, and testing procedures. Sampling, inspecting, and testing procedures shall be in accordance with MIL-I-25645.

4.3 Qualification testing.

4.3.1 Qualification test sample tested by the contractor. The contractor shall subject one indicator set to the qualification tests specified in 4.3.4.

4.3.2 Qualification test report. After the contractor completes the qualification tests, he shall prepare a qualification test report in accordance with MIL-STD-831 and shall furnish three complete copies of the report to the qualifying activity.

4.3.3 Qualification test sample for the qualifying activity. Along with the qualification test report, the contractor shall submit the indicator set that was tested as specified in 4.3.1.

4.3.4 Qualification tests. Qualification tests shall consist of all tests described under Test Methods of MIL-I-25645.

4.3.5 Qualification reliability tests. One qualification sample (one indicator set) shall be tested for reliability and shall be continued through a longevity test in accordance with MIL-STD-781. Acceptance shall be judged on the basis of the accept-reject criteria of Test Plan I of MIL-STD-781. Recording, data handling, and reporting procedures shall be in accordance with MIL-STD-781.

5. PACKAGING.

5.1 Preservation. Preservation shall be Level A, C or commercial/industrial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. The indicators shall be cleaned in accordance with MIL-P-116, process C-1. ANY substance selected for the C-1 cleaning process SHALL NOT be a Class I Ozone Layer Depletion Substance.

5.1.1.2 Drying. The indicators shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservation applications. Preservatives shall not be used.

5.1.1.4 Packaging. Unless otherwise specified (see 6.2) each indicator shall be individually packaged in accordance with MIL-P-116, Method IC. ANY procedure in MIL-P-116J, or earlier revisions, that involves the use of Ozone Layer Depletion Substances shall not be used.

5.1.2 Level C. The Level C preservation for indicators shall conform to the requirements of MIL-STD-2073.

5.1.3 Commercial. The commercial/industrial preservation of indicators shall be in accordance with the requirements of ASTM D 3951.

5.2 Packing. Packing shall be Level A, B, C or commercial, as specified (see 6.2).

5.2.1 Level A. The packaged indicators shall be packed in containers as specified in MIL-STD-2073 for Level A protection. Interior cushioning, blocking, bracing, and waterproofing shall be in accordance with MIL-STD-1186.

5.2.2 Level B. The packaged indicators shall be packed in containers as specified in MIL-STD-2073 for Level B protection.

5.2.3 Level C. The packaged indicators shall be packed in containers as specified in MIL-STD-2073 for Level C protection.

5.2.4 Commercial. Commercial/industrial packing of the indicator shall be paced in accordance with ASTM D 3951.

5.3 Marking.

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5.3.1 Level A, B and C. In addition to any special or other identification marking required by the contract (6.2) each unit pack, intermediate and exterior container and unitized load shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial. Commercial/industrial marking shall be in accordance with the requirements of ASTM D 3951.

6. NOTES

6.1 Intended use. The indicator set covered by this specification is intended for use in indicating the quantity of liquid oxygen remaining in converters in T-38 airplanes.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Levels of preservation/packaging and packing (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in Qualified Products List (QPL No.) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is SA-ALC/TILDD, Kelly Air Force Base, Texas 78241 and information pertaining to qualification of products may be obtained from that activity.

6.4 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

6.5 Subject term (key word) listing.

Cases
Dials
Electrical Connectors
Weight

Custodians:
Air Force - 99

Review activities:
DLA - GS

Preparing activity:
Air Force - 82

(Project No. 6680-F238)

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